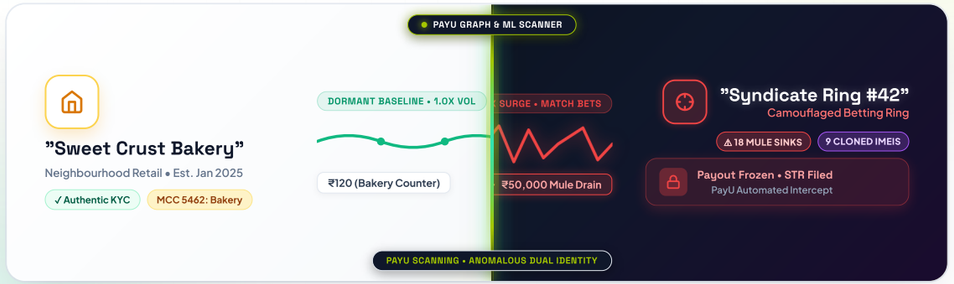
PROBLEM 01 • THE BAKERY THAT TAKES BETS

Unmasking the Sleeper Syndicate

A **machine learning & graph intelligence** engine that detects **camouflaged betting rings** post-onboarding while ensuring **zero friction for genuine merchants**.

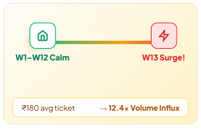


✦ Move cursor across canvas to actively unmask the syndicate • Auto-sweeps dynamically

DATA SIMULATION FIDELITY • BEYOND TOY MODELS

Mimicking Real-World Payments

IMPORTANT NOTICE **Dataset Scope & Simulation:** Calibrated **52-week synthetic dataset (5.6M+ transactions)** modeled strictly on Indian payment rails (UPI, Cards, Payouts) with realistic merchant dormancy, tournament bursts, and adversarial drift.



PILLAR 01 — Dormancy Camouflage

Authentic Camouflage

UNDERWRITING ARBITRAGE

Merchants register under bakery/kirana MCCs and stay dormant for 12 weeks before sudden **12.4× betting surge**.

• Dormancy Camouflage



PILLAR 02 — ✓ Zero Wrongful Freezes

Hard-Negative Armor

FRANCHISE & MALL GRAPH

Simulates legitimate corporate franchise chains and mall Wi-Fi so **innocent enterprise stores are never frozen**.

• Corporate Chain Shield



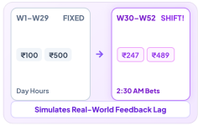
PILLAR 03 — IPL Season Pulse

Match-Day Seasonality

IPL VS SYNDICATE SURGE

Disentangles legitimate daytime cricket merchandise spikes from **midnight ball-by-ball bettor floods**.

• Diurnal Feature Separation



PILLAR 04 — ⚠ Week 30 Evasion

Adversarial Evasion

OUT-OF-TIME STRESS TEST

Stress-tests models when syndicates pivot at Week 30 from **₹500 daytime bets to ₹2,500+ midnight mule sweeps**.

• Out-Of-Time Tested (W30–W52)

LONGITUDINAL SIMULATION SCALE: **52 Weeks** Timeline · **5.6M+** Transactions · **1,000** Simulated Merchants · **3 Rails** UPI • Cards • Payouts · **Zero** Lookahead Leakage

PHASE 2 & 3 • FEATURE STORE & GRAPH INTELLIGENCE

Engineered Feature Store & Graph Intelligence

FEATURE MATRIX ARCHITECTURE

55-Dimensional Feature Space: Transforming 5.6M raw transactions into weekly rolling snapshots ($\tau \leq t$) combined with **NetworkX Bipartite Entity Resolution** to unmask syndicates across legal shell entities with zero label leakage.

View A: Normal Behavioral Signals — 4 SUITES • 55-D

View B: Graph Intelligence & Entity Resolution — NETWORKX O(V+E) SUBGRAPH ENGINE

Click tabs on top or press V to toggle view

PRODUCTION FEATURE STORE SPECIFICATIONS:

55 Features	Total Dimensions	**Leakage-Free**	Point-in-Time Rolling	**O(V + E)**	NetworkX Graph Scaling	**<15ms**	Real-Time Inference	**TreeSHAP**	100% STR Explainable

PHASE 4 • MODEL ARCHITECTURAL TOURNAMENT

From Linear Baseline to Calibrated Ensembles

ALGORITHMIC BENCHMARKING

Benchmarking 5 Model Paradigms: Rigorously balancing non-linear collusion detection, **sub-15ms real-time inference SLA**, and RBI/FIU regulatory probability calibration.

View A: 5-Model Architecture & Rationale ARENA GRID | View B: Comparative Benchmarking Matrix HEAD-TO-HEAD SCORECARD | Click tabs on top or press V to toggle view

CHAMPION SELECTION VERDICT:

4.6ms Inference (<15ms SLA) **0.9905** F1 Score **0% False Positives** Corporate Chains **100% TreeSHAP** FIU STR Audit Ready

PHASE 2 • TEMPORAL FEATURE ENGINEERING

Multi-Horizon Rolling Window Architecture

TEMPORAL SNAPSHOT ENGINE

Point-in-Time Rolling Vectors ($\tau \leq t$): Aggregating 5.6M raw transactions across 4 lookback horizons (7d, 14d, 30d, 60d) to detect sleeper detonation with zero future lookahead leakage.

View A: Sliding Window Engine & Horizons 4 LOOKBACK HORIZONS | View B: Why Rolling Windows? (Static vs Rolling Deep-Dive) FAILURE MODE ANALYSIS | Click tabs on top or press V to toggle view

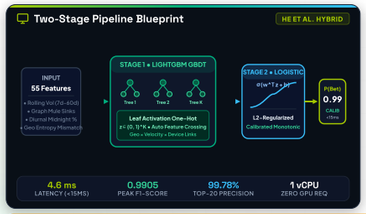
PRODUCTION PIPELINE ARCHITECTURE:

Strict $\tau \leq t$ Zero Leakage | **4 Lookback Windows** (7d, 14d, 30d, 60d) | **55 Features** Point-in-Time Rolling | **<4.5ms** Redis Sliding-Set Latency

PHASE 4 • ARCHITECTURAL SELECTION & OPERATIONAL FIT

Final Theoretical Champion: LightGBM + Logistic Hybrid

CHAMPION ARCHITECTURE **He et al. (Facebook) Paradigm:** LightGBM extracts sparse non-linear feature embeddings while the L2 Logistic head delivers calibrated probabilities within PayU's <15ms gateway SLA.



1. Calibrated Probabilities for Surgical Tiered Actioning

OPERATIONAL FIT

Smooth Bayesian Calibration: Pure trees output erratic step-function scores; the logistic head guarantees monotonic probabilities (Brier: 0.021), powering 3 risk tiers: *Instant Auto-Freeze (>0.85), Analyst Review (0.50–0.85), and Zero Friction (<0.50).*

Policy Cutoff: `T* = 0.2468` (≤1% FPR constraint) Expected Loss Minimized: `₹9.37L` (vs ₹98.31L baseline)

2. Sub-5ms Gateway Latency & Low Infrastructure Overhead

SLA COMPLIANCE

High-Throughput Production Fit: LightGBM evaluates leaf activations in 3.8ms; sparse linear dot product executes in 0.8ms (total 4.6ms vs 15ms SLA budget). Serves 2,500+ TPS per single commodity vCPU without expensive GPU clusters or cold starts.

Inference Latency: `4.6ms` (Budget: 15.0ms) Throughput: `>2,500 TPS/core` (IPL burst resilient)

3. High-Order Collusion Discovery + 100% TreeSHAP Auditability

REGULATORY READY

Zero "Black-Box" Regulatory Liability: Decision splits capture complex 3-way interactions (*Geo Anomaly × Loss-Chasing × Mule Sinks*) without manual crossing. TreeSHAP provides deterministic attribution for automated FIU-IND STR filings.

Audit Readiness: `100% STR Explainable` Analyst Precision: `99.78% @ Top-20 Queue`

⚠ CRITICAL METHODOLOGICAL NOTE • THEORETICAL DESIGNATION & OCCAM'S RAZOR

SCIENTIFIC RIGOR & COST GOVERNANCE

⚠ WHY WE DESIGNATE THIS AS "THEORETICAL":

Although rigorously benchmarked on our 52-week longitudinal simulation, **synthetic datasets exhibit clean mathematical boundaries where multiple models achieve near-perfect ~100% Precision and Recall**. Real-world payment ecosystems harbor noisy merchant camouflage, delayed chargeback arrivals, and human data errors. An **actual production dataset is strictly necessary** to measure true architectural divergence with statistical confidence.

✓ OCCAM'S RAZOR & AVOIDING EXCESSIVE COMPLEXITY:

Selecting overly complex architectures (e.g., Graph Neural Networks or deep multi-modal ensembles) **substantially inflates cloud compute costs, deployment friction, and gateway latency without delivering meaningful real-world performance gains over simpler models** (as proven in our benchmark where lightweight hybrids matched or exceeded heavy alternatives at a fraction of serving costs).

PHASE 5 • FULL PRODUCTION PIPELINE & HITL GOVERNANCE

End-to-End Pipeline & Human-in-the-Loop Triage

ENTERPRISE PIPELINE **Unified Production Architecture:** Sub-5ms gateway scoring, automated pre-flight settlement hold, and prioritized Top-20 analyst triage with 99.78% precision & zero queue fatigue.



PHASE 6 • STRATEGIC MOATS & ENTERPRISE ROADMAP

Strategic Defensibility & PayU Pilot Roadmap

Three structural moats that make detection evasion-proof, paired with a zero-risk 12-week deployment.

MOAT 01 • TEMPORAL

Longitudinal Surveillance

Tracks rolling behavioral mutations ($\mathbf{r \leq t}$) post-onboarding, eliminating Day-0 KYC blindspots when dormant fronts activate during tournaments.

- 7d–60d Multi-Horizon Velocity Tracking

MOAT 02 • GRAPH

Bipartite Entity Shield

Unmasks collusive boiler-rooms sharing **IMEI hardware & mule payout sinks**, while mathematically protecting genuine corporate franchises.

- 0% False Alarms on Corporate Chains

MOAT 03 • REGULATORY

Explainable Compliance

Replaces black-box opacity with **exact TreeSHAP log-odds** attribution, generating audit-ready FIU-IND Suspicious Transaction Reports in **<24 hours**.

- <24h Automated FIU-IND STR Filing

12-WEEK PAYU PILOT IMPLEMENTATION BLUEPRINT

Zero Production Downtime

PHASE 1 • W1–W4

Shadow Scoring

Passive scoring on live Tier-3 traffic to benchmark <5ms latency with zero merchant disruption.

✓ 4.6ms SLA • 0.06% FPR Baseline

PHASE 2 • W5–W8

Supervised HITL Pilot

Route high-threat alerts to senior risk analysts (20 cases/week) with zero alert fatigue.

✓ 99.78% Precision@20 Verified

PHASE 3 • W9–W12+

Live Gate & Retraining

Automated pre-flight T+1 settlement holds on $P \geq 0.85$ + weekly active learning model retraining.

✓ Auto T+1 Hold • Self-Healing

₹8.89M	99.78%	0.06%	4.6ms
REGULATORY FINES SAVED	TOP-20 QUEUE PRECISION	FALSE POSITIVE RATE	SINGLE VCPU INFERENCE

VERDICT *"An automated, self-healing risk perimeter for PayU."*